

**B.Sc. BOTANY  
SYLLABUS**  
(Common Core Syllabus under CBCS)

**with effect from the academic year 2025-2026 and  
onwards**



*Accredited with 'A+' by NAAC*

**DEPARTMENT OF BOTANY  
KAKATIYA UNIVERSITY  
WARANGAL -506 009, TELANGANA**

**KAKATIYA UNIVERSITY**  
**WARANGAL-506009, TELANGANA**  
 (COMMON CORE SYLLABUS UNDER CBCS FOR)

**B. Sc BOTANY**

| CODE                           | PAPER/TITLE                                                                                                   | Course Type                  | HPW     | Credits |
|--------------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------|---------|---------|
| <b>FIRST YEAR SEMSTER-I</b>    |                                                                                                               |                              |         |         |
| BS104                          | PAPER-I: Microbial Diversity and Early Land Plants                                                            | DSC-1A                       | 4T+2P=6 | 4+1=5   |
| <b>FIRST YEAR SEMSTER-II</b>   |                                                                                                               |                              |         |         |
| BS204                          | PAPER-II: Gymnosperms, Anatomy and Embryology of Angiosperms                                                  | DSC-1B                       | 4T+2P=6 | 4+1=5   |
| <b>SECOND YEAR SEMSTER-III</b> |                                                                                                               |                              |         |         |
| BS302                          | PAPER-III: Plant Taxonomy, Ecology and Medicinal Botany                                                       | DSC-1C                       | 4T+2P=6 | 4+1=5   |
| <b>SECOND YEAR SEMSTER-IV</b>  |                                                                                                               |                              |         |         |
| BS402                          | PAPER-IV: Cell Biology, Genetics & Plant Physiology                                                           | DSC-1D                       | 4T+2P=6 | 4+1=5   |
| <b>THIRD YEAR SEMESTER-V</b>   |                                                                                                               |                              |         |         |
| BS501                          | SEC:1-                                                                                                        | SEC-1                        | 2       | 2       |
| BS502                          | SEC:2-                                                                                                        | SEC-2-                       | 2       | 2       |
| BS503                          | VAC-1:                                                                                                        | VAC-1                        | 3       | 3       |
| BS504                          | Multi-Disciplinary Course (MDC)                                                                               | MDC                          | 4T      | 4       |
| BS505                          | DSE-1A: Biodiversity & Conservation<br>DSE-1B: Tissue Culture and Biotechnology<br>DSE-1C: Economic Botany    | DSE-1A/<br>DSE-1B/<br>DSE-1C | 4T+2P=6 | 4+1=5   |
| <b>THIRD YEAR SEMESTER-VI</b>  |                                                                                                               |                              |         |         |
| BS601                          | SEC:3                                                                                                         | SEC:3-                       | 2       | 2       |
| BS602                          | SEC:4                                                                                                         | SEC:4-                       | 2       | 2       |
| BS603                          | VAC-2                                                                                                         | VAC-2                        | 3       | 3       |
| BS604                          | DSE-2A: Plant Molecular Biology<br>DSE-2B: Seed Technology<br>DSE-2C: Analytical Techniques in Plant Sciences | DSE-2A/<br>DSE-2B/<br>DSE-2C | 4T+2P=6 | 4+1=5   |
|                                | PROJECT                                                                                                       |                              |         | 4       |

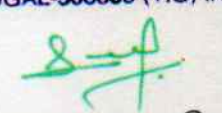
SEC: Skill Enhancement Course, VAC: Value added Course, MDC: Multi-Disciplinary Course  
 DSC: Discipline Specific Core, DSE: Discipline Specific Elective.

  
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**Dr. M. Mahamatha, Ph.D**  
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**B. Lalitha Kumari**

**B.Sc. BOTANY**  
**I Year: I-Semester**  
**Paper-I: Microbial Diversity and Early Land Plants**

**DSC-1A**

**(4hrs./week)**

**Credits-4**

**Theory Syllabus**

**(60hours)**

**UNIT-I**

**(15hours)**

- 1) Brief account of Archaeobacteria, Actinomycetes and Mycoplasma with reference to Little leaf of Brinjal and Papaya leaf curl.
- 2) **Viruses:** Structure, replication and transmission; plant diseases caused by viruses and their control with reference to Tobacco Mosaic and Rice Tungro.
- 3) **Bacteria:** Structure, nutrition and reproduction. Plant diseases caused by bacteria and their control with reference to Angular leaf spot of Cotton and Bacterial blight of Rice.

**UNIT-II**

**(15hours)**

- 4) General characters, structure, reproduction and classification of Algae (Fritsch)
- 5) **Cyanobacteria:** General characters, cell structure their significance as biofertilizers with special reference to *Oscillatoria*, *Nostoc* and *Anabaena*.
- 6) Structure and reproduction of the following:  
Chlorophyceae - *Volvox*, and *Chara*.  
Phaeophyceae - *Ectocarpus*  
Rhodophyceae - *Polysiphonia*.

**UNIT-III**

**(15hours)**

- 7) General characters and classification of fungi (Ainsworth).
- 8) Structure, reproduction and life cycle of the following:  
(a) Mastigomycotina - *Albugo*  
(b) Zygomycotina - *Mucor*  
(c) Ascomycotina - *Penicillium*.  
(d) Basidiomycotina - *Puccinia*  
(e) Deuteromycotina - *Cercospora*.
- 9) Economic importance of Lichens

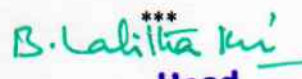
**UNIT-IV**

**(15hours)**

- 10) **Bryophytes:** Structure, reproduction, life cycle and systematic position of *Marchantia*, and *Polytrichum*, Evolution of Sporophyte in Bryophytes.
- 11) **Pteridophytes:** Structure, reproduction, life cycle and systematic position of *Rhynia*, *Equisetum* and *Marsilea*.
- 12) Stellar evolution, heterospory and seed habit in Pteridophytes.

  
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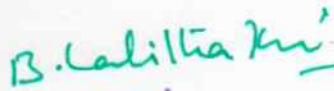
  
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### Suggested Readings:

- 1) Alexopolous, J. and W.M. Charles. 1988. Introduction to Mycology. Wiley Eastern, New Delhi.
- 2) Mckane, L. and K. Judy. 1996. Microbiology—Essentials and Applications. Mc Graw Hill, New York.
- 3) Pandey, B.P. 2001. College Botany, Vol. I: Algae, Fungi, Lichens, Bacteria, Viruses, Plant Pathology, Industrial Microbiology and Bryophyta. S. Chand & Company Ltd, New Delhi.
- 4) Pandey, B.P. 2007. Botany for Degree Students: Diversity of Microbes, Cryptogams, Cell Biology and Genetics. S. Chand & Company Ltd, New Delhi.
- 5) Sambamurthy, A.V.S.S. 2006. A Textbook of Plant Pathology. I.K. International Pvt. Ltd., New Delhi.
- 6) Sambamurthy, A.V.S.S. 2006. A Text book of Algae. I.K. International Pvt. Ltd., New Delhi.
- 7) Sharma, O.P. 1992. Text book of Thallophyta. McGraw Hill Publishing Co., New Delhi.
- 8) Thakur, A.K. and S.K. Bassi. 2008. A Textbook of Botany: Diversity of Microbes and Cryptogams. S. Chand & Company Ltd, New Delhi.
- 9) Vashishta, B.R., A.K. Sinha and V.P. Singh. 2008. Botany for Degree Students: Algae. S. Chand & Company Ltd, New Delhi.
- 10) Vashishta, B.R. 1990. Botany for Degree Students: Fungi, S. Chand & Company Ltd, New Delhi.
- 11) Dutta A.C. 2016. Botany for Degree Students. Oxford University Press.
- 12) Watson, E.V. 1974. The structure and life of Bryophytes, B.I. Publications, New Delhi.
- 13) Pandey, B.P. 2006. College Botany, Vol. II: Pteridophyta, Gymnosperms and Paleobotany. S. Chand & Company Ltd, New Delhi.
- 14) Vashishta, P.C., A.K. Sinha and Anil Kumar. 2006. Botany-Pteridophyta (Vascular Cryptogams). S. Chand & Company Ltd, New Delhi.
- 15) Pandey, B.P. 2001. College Botany, Vol. I: Algae, Fungi, Lichens, Bacteria, Viruses, Plant Pathology, Industrial Microbiology and Bryophyta. S. Chand & Company Ltd, New Delhi.
- 16) Pandey, B.P. 2007. Botany for Degree Students: Diversity of Microbes, Cryptogams, Cell Biology and Genetics. S. Chand & Company Ltd, New Delhi.
- 17) Thakur, A.K. and S.K. Bassi. 2008. A Textbook of Botany: Diversity of Microbes and Cryptogams. S. Chand & Company Ltd, New Delhi.
- 18) Vashishta, B.R., A.K. Sinha and Adarsha Kumar. 2008. Botany for Degree Students: Bryophyta. S. Chand & Company Ltd, New Delhi.

  
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**B.Sc.BOTANY**  
**I Year: I-Semester**  
**Paper-I: Microbial Diversity and Early Land Plants**  
**DSC-1A Credits-1**

**Practical Syllabus**

**(45hours)**

1. Study of viruses and bacteria using electron micrographs(photographs).
2. Gram staining of Bacteria.
3. Study of symptoms of plant diseases caused by viruses, bacteria, Mycoplasma and fungi: **Viruses:** Tobacco mosaic and Rice tungro; **Bacteria:** Angular leaf spot of Cotton and Bacterial blight of Rice; **Mycoplasma:** Little leaf of Brinjal and Leaf curl of Papaya; **Fungi:** White rust on Crucifers, Rust on Wheat & Tikka disease of Groundnut.
4. Vegetative and reproductive structures of the following taxa:  
**Algae:** *Oscillatoria*, *Nostoc*, *Volvox*, *Chara*, *Ectocarpus* and Polysiphonia.  
**Fungi:** *Albugo*, *Mucor*, *Penicillium*, *Puccinia* and *Cercospora*
5. Section cutting of diseased material infected by Fungi and identification of pathogens as per theory syllabus. (White rust of Crucifers, Rust on Wheat & Tikka disease of Groundnut).
6. Study of vegetative, reproductive structures and anatomy of **Bryophytes:** *Marchantia*, and *Polytrichum*.
7. Study of vegetative, reproductive structures and anatomy of **Pteridophytes:** *Equisetum* and *Marsilea*.
8. Study of Anatomical features of *Equisetum* stem and *Marsilea* Petiole & Rhizome by preparing temporary mounts.
9. Field visits to places for algal / microbial / fungal interest(e.g. Mushroom cultivation, algal collection and infected plant materials).

**Practical Model Paper**

**Max.Marks:25**

**Time: 2hrs**

- I. Identify the given components 'A' & 'B' in the algal mixture.  
Describe with neat labeled diagrams & give reasons for the classifications. 2X2=4M
- II. Classify the given bacterial culture 'C' using Gram s staining technique. 3M
- III. Take a thin transverse section of given diseased plant material / Pteridophyte material 'D'.  
Identify & describe the symptoms caused by the pathogen / describe with neat labeled diagram. 4M
- IV. Identify the given diseased plant material specimens 'E', 'F' & 'G' by giving reasons.  
(Fungal-1, Bacteria-1 & Viral/Mycoplasma-1) 3X1=3M
- V. Comment on the given slides 'H' & 'I' (Algae-1, Fungi-1) 2X2=4M
- VI. Identify the given specimen / slide 'J' & 'K' (Bryophytes -1 & Pteridophytes -1) 2X2=4M
- VII. Record & Viva 3M

  
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**B. Lakshmi**

**B.Sc., BOTANY**  
**I Year, II-Semester**  
**Paper-II. Gymnosperms, Anatomy and Embryology of Angiosperms**

**DSC-1B** (4hrs./week) Credits-4

**Theory Syllabus** (60hours)

**UNIT-I** (15hours)

1. Gymnosperms: Distribution, General characters, structure, reproduction and classification (Sporne, 1965)  
Economic importance of Gymnosperms.
2. Morphology of vegetative and reproductive parts, systematic position and life cycle of *Pinus* and *Gnetum*.
3. Introduction to Palaeobotany, Types of fossils and fossilization, Importance of fossils.

**UNIT-II** (15h)

4. Meristems: Types, histological organization of shoot and root apices and theories.
5. Tissues and Tissue systems: Simple, complex and special tissues.
6. Leaf: Internal structure of dicot and monocot leaf. Stomata structure and types. Epidermal out growths.

**UNIT-III** (15h)

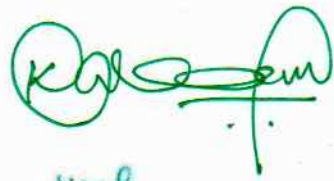
7. Secondary Growth: Vascular cambium – structure and function, Secondary growth in root and stem, Wood (heartwood and sapwood).
8. Anomalous secondary growth of Stem - *Achyranthes*, *Boerhaavia*, *Dracaena*; Root – *Beta*
9. Wood structure: General account. Study of local timbers – Teak (*Tectona grandis*), Red sanders (*Pterocarpus santalinus*) and Neem (*Azadirachta indica*).

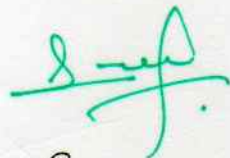
**UNIT-IV** (15h)

10. Structure of Anther, Microsporogenesis and development of male gametophyte.
11. Ovule structure and types; Megasporogenesis and development of female gametophyte.
12. Pollination mechanisms, Pollen-pistil interaction; Double fertilization
13. Types of Endosperm. Embryo structure – Dicot and Monocot. Polyembryony and Apomixis –an outline.

  
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**A. Subba Rao**

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2. Pandey, B.P. 2006. College Botany, Vol. II: Pteridophyta, Gymnosperms and Paleobotany. S. Chand & Company Ltd, New Delhi.
3. Sporne, K.R. 1965. Morphology of Gymnosperms. Hutchinson Co., Ltd., London.
4. Vashishta, P.C., A.K. Sinha and Anil Kumar. 2006. Botany-Pteridophyta (Vascular Cryptogams). S. Chand & Company Ltd, New Delhi.
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11. Dutta A.C. 2016. Botany for Degree Students. Oxford University Press.
12. Pandey, B.P. 2007. Botany for Degree Students: Diversity of Seed plants and their Systematics, Structure, Development and Reproduction in Flowering Plants. S. Chand & Company Ltd, New Delhi.
13. Bhattacharya et al. 2007. A text book of Palynology, Central, New Delhi.
14. Bhojwani, S.S. and S.P. Bhatnagar. 2000. The Embryology of Angiosperms (4th Ed.), Vikas Publishing House, Delhi.
15. M.R. Saxena - A text book of Palynology. 4. Vashista - A text book of Anatomy.
16. P.K.K. Nair - A text book of Palynology
17. Evert, R.F. (2006) Esau's Plant Anatomy: Meristems, Cells, and Tissues of the Plant Body: Their Structure, Function and Development. John Wiley and Sons, Inc.
18. Esau, K. 1971. Anatomy of Seed Plants. John Wiley and Son, USA.
19. Johri, B.M. 1984. Embryology of Angiosperms. Springer-Verlag, Berlin.
20. Kapil, R.P. 1986. Pollination Biology. Inter India Publishers, New Delhi.
21. Maheswari, P. 1971. An Introduction to Embryology of Angiosperms. McGraw Hill Book Co., London.
22. Dutta A.C. 2016. Botany for Degree Students. Oxford University Press.
23. Bhojwani, S.S. and Bhatnagar, S.P. (2011). The Embryology of Angiosperms, Vikas Publishing House, Delhi. 5th edition
24. Shivanna, K.R. (2003). Pollen Biology and Biotechnology. Oxford and IBH Publishing Co. Pvt. Ltd. Delhi.
25. Raghavan, V. (2000). Developmental Biology of Flowering plants, Springer, Netherlands. 4. Johri, B.M. (1984). Embryology of Angiosperms, Springer-Verlag, Netherlands.

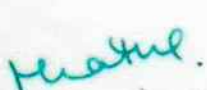
  
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**I Year,II-Semester**  
**Paper-II.Gymnosperms,Anatomy and Embryology of Angiosperms**

**DSC-1B**

**Credit 1**

**Practical Syllabus**

**(45 hours)**

1. Study of vegetative and reproductive structures of the following taxa:  
Gymnosperms - *Pinus* and *Gnetum*.
2. Demonstration of double staining technique.
3. Study of Anatomical features of *Pinus* needle and *Gnetum* Stem by preparing double stained permanent mounts.
4. Fossil forms using permanent slides / photographs: Cycadeoidea.
5. Tissue organization in root and shoot apices using permanent slides.
6. Study of different tissue systems-Simple, complex and special tissues
7. Preparation of double stained Permanent slides of Anomalous secondary structure:  
Stem - *Achyranthes*, *Boerhaavia*, *Dracaena*; Root-*Beta*
8. Stomatal types using epidermal peels.
9. Structure of anther and microsporogenesis using permanent slides.
10. Structure of pollen grains using whole mounts-*Hibiscus*, *Acacia* and *Grass*
11. Pollen viability test using Evans Blue
12. Study of ovule types and developmental stages of embryo sac.
13. Structure of endosperm (nuclear and cellular).
14. Developmental stages of dicot and monocot embryos using permanent slides.
15. Isolation and mounting of embryo (using *Cyamopsis*/*Crotalaria*)

**Practical Model Paper**

**Time: 2hrs**

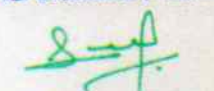
**Max.Marks:25**

1. Prepare a double stained permanent mount of the given material "A" (Gymnosperms) 5M  
Draw diagram & give reasons for identification.
2. Identify the given material "B" (Anomalous secondary growth). 4M  
Prepare a double stained permanent mount of transverse section.
3. Prepare a temporary mount of epidermal peel of the given leaf material "C" and 3M  
Identify the stomatal type.
4. Conduct the pollen viability test (OR) Isolate the embryo from the given material "D" 3M  
Identify and describe the specimens / slides with well labeled diagrams  
(a) Gymnosperms-E (b) Anatomy-F (c) Embryology- G (3X2)=6M
5. Record and Viva 4M

  
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**Annexure – I (Credits)**  
**Proposed CBCS Structure from 2025-26 for Under Graduate Courses**

| Courses                                                         |                         | Papers                                       | Total Credits | Credits for each paper / Semester      |    |     |    |    |    | Credits for each paper / Semester      |    |     |    |    |    | Credits for each paper / Semester      |    |     |    |    |    |
|-----------------------------------------------------------------|-------------------------|----------------------------------------------|---------------|----------------------------------------|----|-----|----|----|----|----------------------------------------|----|-----|----|----|----|----------------------------------------|----|-----|----|----|----|
|                                                                 |                         |                                              |               | BA                                     |    |     |    |    |    | B.Com.                                 |    |     |    |    |    | B.Sc.                                  |    |     |    |    |    |
|                                                                 |                         |                                              |               | I                                      | II | III | IV | V  | VI | I                                      | II | III | IV | V  | VI | I                                      | II | III | IV | V  | VI |
| Core Courses<br>DSC                                             | Major-1                 | 6                                            | 30            | 5                                      | 5  | 5   | 5  | 5  | 5  | 5                                      | 5  | 5   | 5  | 5  | 5  | 5                                      | 5  | 5   | 5  | 5  | 5  |
|                                                                 | Major -2                | 6                                            | 30            | 5                                      | 5  | 5   | 5  | 5  | 5  | 5                                      | 5  | 5   | 5  | 5  | 5  | 5                                      | 5  | 5   | 5  | 5  | 5  |
|                                                                 | Minor-1                 | 4                                            | 20            | 5                                      | 5  | 5   | 5  | -  | -  | 5                                      | 5  | 5   | 5  | -  | -  | 5                                      | 5  | 5   | 5  | -  | -  |
| MIL/AEC<br>(First Language)                                     | English                 | 4                                            | 20            | 5                                      | 5  | 5   | 5  | -  | -  | 5                                      | 5  | 5   | 5  | -  | -  | 5                                      | 5  | 5   | 5  | -  | -  |
| Second Language<br>(Telugu, Hindi, Urdu, etc.)                  |                         | 4                                            | 20            | 5                                      | 5  | 5   | 5  | -  | -  | 5                                      | 5  | 5   | 5  | -  | -  | 5                                      | 5  | 5   | 5  | -  | -  |
| Multi-<br>Disciplinary<br>Course                                | MDC 1                   | 1                                            | 4             | -                                      | -  | -   | -  | 4  | -  | -                                      | -  | -   | -  | 4  | -  | -                                      | -  | -   | -  | 4  | -  |
| Sec 1, 2                                                        |                         | 2                                            | 4             |                                        |    |     |    | 2  | 2  |                                        |    |     |    | 2  | 2  |                                        |    |     |    | 2  | 2  |
| Sec 3, 4                                                        |                         | 2                                            | 4             |                                        |    |     |    | 2  | 2  |                                        |    |     |    | 2  | 2  |                                        |    |     |    | 2  | 2  |
| Value added<br>course (VAC)                                     | VAC 1, 2                | 2                                            | 6             | -                                      | -  | -   | -  | 3  | 3  | -                                      | -  | -   | -  | 3  | 3  | -                                      | -  | -   | -  | 3  | 3  |
| Internships                                                     | Internship<br>/ Project | 1                                            | 4             | -                                      | -  | -   | -  | -  | 4  | -                                      | -  | -   | -  | -  | 4  | -                                      | -  | -   | -  | -  | 4  |
| Total Credits in each semester                                  |                         |                                              | 142           | 25                                     | 25 | 25  | 25 | 21 | 21 | 25                                     | 25 | 25  | 25 | 21 | 21 | 25                                     | 25 | 25  | 25 | 21 | 21 |
| Total Credits in UG                                             |                         |                                              |               | 142                                    |    |     |    |    |    | 142                                    |    |     |    |    |    | 142                                    |    |     |    |    |    |
| Credits under Non-CGPA<br>(Community engagement and<br>service) |                         | NSS /NCC<br>/sports /<br>Extra<br>curricular | 6             | Upto 6 (2 in each year)                |    |     |    |    |    | Upto 6 (2 in each year)                |    |     |    |    |    | Upto 6 (2 in each year)                |    |     |    |    |    |
|                                                                 |                         | IKS                                          | 4             | Upto 4 (2 in each, after I & II years) |    |     |    |    |    | Upto 4 (2 in each, after I & II years) |    |     |    |    |    | Upto 4 (2 in each, after I & II years) |    |     |    |    |    |



**B.Sc. BOTANY  
SYLLABUS**  
(Common Core Syllabus under CBCS)

**with effect from the academic year 2026-2027 and  
onwards**



*Accredited with 'A+' by NAAC*

**DEPARTMENT OF BOTANY  
KAKATIYA UNIVERISTY  
WARANGAL -506 009, TELANGANA**

**Department of Botany**  
**Kakatiya University, Warangal – 506 009**  
**Telangana State**

Meeting of the Board of Studies in Botany was held on **18-06-2026 at 11.00 a.m.** in Head's Chamber, Department of Botany, Kakatiya University, Warangal.

**Agenda:**

1. To discuss and approval of UG B.Sc. Botany Semester III & IV revised syllabus given by TGCHE.
2. Any necessary modifications to be made in Theory and Practical syllabus of Semester III & IV.

**The following members were present:**

- |                                                                                       |   |                 |
|---------------------------------------------------------------------------------------|---|-----------------|
| 1. The Chairperson, Board of Studies in Botany<br>(Dr. P. Karunakar Rao)              | - | Chairperson     |
| 2. The I/c Head, Department of Botany, KU<br>(Dr. K. Mamatha)                         | - | Member          |
| 3. Dr. K. Saritha<br>Asst. Professor, GDC, Mulugu                                     | - | Member          |
| 4. Dr. K. Omkar<br>Associate Professor, SR&BGNR Government College, Khammam           | - | Member          |
| 5. Prof. A. Sabitha Rani<br>Head, Dept. of Botany, Osmania University, Hyderabad.     | - | External Member |
| 6. Prof. B. Ravi Prasad Rao<br>Head, Dept. of Botany, S.K. University, Ananthapur, AP | - | External Member |
| 7. Prof. M. Mamatha<br>Forest Botany, Forest College and Research Institute, Mulugu   | - | External Member |

**Resolutions:**

1. It is resolved to approve the revised syllabus of UG B.Sc. Botany for III & IV semesters (Theory & Practical) given by the TGCHE with effect from 2026-2027 academic year.
2. It is resolved to modify the practical syllabus of Semesters III & IV slightly.
3. It is also resolved to approve the revised syllabus of UG B.Sc. Agriculture / Crop Production for III & IV semesters.

  
**Head**  
Dept. of Botany, Kakatiya University  
Warangal, Telangana-506009

  
**DR. P. KARUNAKAR RAO**  
Chairperson  
Board of Studies in Botany  
Kakatiya University,  
WARANGAL-506009 (T.G.)


**B.Sc., BOTANY**  
**II Year, III -Semester**

**Paper- III. Plant Taxonomy, Ecology and Medicinal Botany**

DSC - 1C (4 hrs./week)

Credits- 4

**Theory Syllabus**

**UNIT-I**

(60 hours)

- (15 h)
1. Introduction: History, Principles, objectives and opportunities in plant taxonomy/ plant systematics.
  2. Systems of classification: Salient features and comparative account of Linnaeus, Bentham & Hooker and Engler & Prantl classification systems. An introduction to Angiosperm Phylogeny Group (APG).
  3. Current concepts in Angiosperm Taxonomy: Embryology in relation to taxonomy, Cytotaxonomy and Chemotaxonomy.
  4. Nomenclature and Taxonomic resources: An introduction to ICN. Herbarium: Concept, techniques and applications. Role of botanical gardens in Taxonomic research.

**UNIT-II**

- (15h)
5. Systematic study and economic importance of plants belonging to the following families:  
Polypetalae: Annonaceae, Rutaceae, Cucurbitaceae
  6. Gamopetalae: Asteraceae, Asclepiadaceae, Lamiaceae
  7. Monochalmydeae: Euphorbiaceae
  8. Monocotyledons: Orchidaceae and Poaceae.

**UNIT III.**

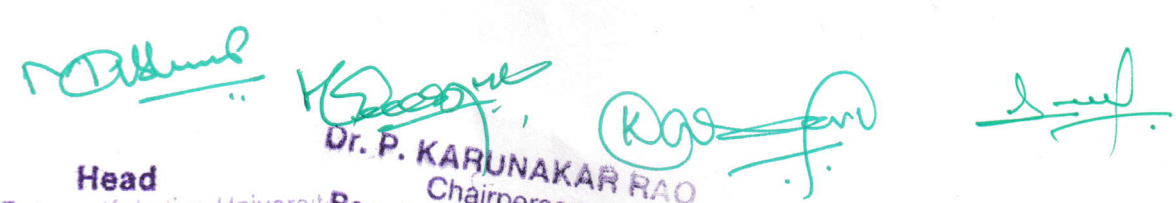
(15h)

9. Component of Ecosystem, energy flow, food chain and food webs.
10. Plant population studies- - Abundance, density and Frequency of vegetation by Quadrat Method
11. Plants and environment: Ecological adaptations of plants, Hydrophytes and Xerophytes.
12. Plant Succession: Serial stages, climax formation with reference to Hydrosere and Xerosere.

**Unit IV**

(15h)

13. Introduction: History, scope and interdisciplinary nature of medicinal botany. Ethnomedicine and folklore medicine.
14. Traditional medicinal systems- basic principles, methods of preparation of medicines in Ayurveda, Siddha, Unani and Homeopathy.
15. Plants used in primary health care: Study of morphology and traditional uses of following: Black pepper (*Piper nigrum*), Garlic (*Allium sativum*), Ginger (*Zingiber officinale*) Turmeric (*Curcuma longa*).
16. Plants used in traditional medicine: Study of morphology, active principles, medicinal uses of following: Amla (*Phyllanthus emblica*), Ashwagandha (*Withania somnifera*), Podapatri (*Gymnema sylvestre*), Sarpagandha (*Rauvolfia serpentina*), Tippa teega (*Tinospora cordifolia*).

  
**Head**  
Dept. of Botany, Kakatiya University  
Warangal, Telangana-506009

**Dr. P. KARUNAKAR RAO**  
Chairperson  
Board of Studies in Botany  
Kakatiya University,  
WARANGAL-506009 (I.G) II

### Suggested Readings:

1. Stace, C. A. 1989. Plant Taxonomy and Biostatistics (2nd Ed.). Edward Arnold, London.
2. Singh, G. 1999. Plant Systematics: Theory and Practice. Oxford and IBH, New Delhi.
3. Dutta A.C. 2016. Botany for Degree Students. Oxford University Press.
4. Davis, P. H. and V. H. Heywood. 1963. Principles of Angiosperm Taxonomy. Oliver and Boyd, London.
5. Heywood, V. H. 1965 . Plant Taxonomy. ELBS , London.
6. Heywood, V. H. and D. M. Moore (Eds). 1984. Current Concepts in Plant Taxonomy. Academic Press, London.
7. Jeffrey, C. 1982. An Introduction to Plant Taxonomy. Cambridge University Press, Cambridge.London
8. Singh, G. (2012). Plant Systematics: Theory and Practice Oxford & IBH Pvt. Ltd., New Delhi. 2. 3rd edition.
9. Jeffrey, C. (1982). An Introduction to Plant Taxonomy. Cambridge University Press, Cambridge.
10. Judd, W.S., Campbell, C.S., Kellogg, E.A., Stevens, P.F. (2002). Plant Systematics-A Phylogenetic Approach. Sinauer Associates Inc., U.S.A. 2nd edition.
11. Radford, A.E. (1986). Fundamentals of Plant Systematics. Harper and Row, New York
12. Simpson, M. G. (2006). Plant Systematics. Elsevier. 7. Cooke, T. (1903). The Flora of Presidency of Bombay, I, London, (B.S.I. Repr. ed., 1958.) Botanical Survey of India, Calcutta
13. Kormondy, E.J. (1996). Concepts of Ecology. Prentice Hall, U.S.A. 4th edition.
14. Sharma, P.D. (2010) Ecology and Environment. Rastogi Publications, Meerut, India. 8th edition.\
15. Odum, E.P. (2005). Fundamentals of ecology. Cengage Learning India Pvt. Ltd., New Delhi. 5th edition.
16. Singh, J.S., Singh, S.P., Gupta, S. (2006). Ecology Environment and Resource Conservation. Anamaya Publications, New Delhi, India
17. Wilkinson, D.M. (2007). Fundamental Processes in Ecology: An Earth Systems Approach. Oxford University Press. U.S.A
18. Michael, S. 1996, Ecology, Oxford University Press, London14



**Head**  
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**Dr. P. KARUNAKAR** M.A.  
Chairperson  
Board of Studies in Botany  
Kakatiya University,  
WARANGAL-506006 (T.G) INDIA

**B.Sc., BOTANY**  
**II Year, III -Semester**

**Paper- III. Plant Taxonomy, Ecology and Medicinal Botany**

DSC - 1C

(Credits- 1)

**Practical Syllabus**

(45 hours)

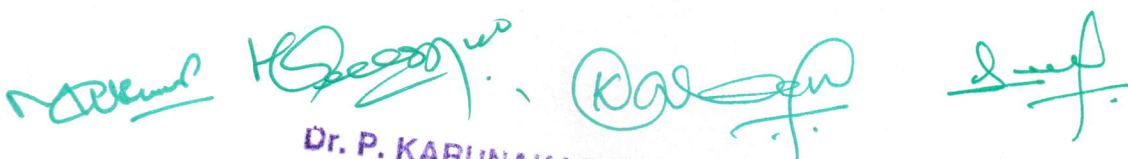
1. Systematic study of locally available plants belonging to the families prescribed in theory Syllabus (Minimum of one plant representative for each family)
2. Demonstration of herbarium techniques.
3. Study of morphological and anatomical characteristics of locally available plant species (*Eichhorinia*, *Hydrilla*, *Pistia*, *Nymphaea*, *Asparagus*, *Opuntia*, *Euphorbia melii*).
4. Calculation of density, abundance and frequency of vegetation in the campus/ nearby areas by Quadrature method.
5. Study of morphology and traditional uses of plant used in primary health care (prescribed in the theory Syllabus)
6. Study of the plants used in traditional medicine (prescribed in the theory syllabus)
7. Candidate has to submit at least 30 herbarium sheets

**Practical Model Paper**

**Time: 3 hrs**

**Max. Marks: 25**

1. Identify and give a technical description of the plant twig - 'A' 7 M
2. Calculation of density, abundance and frequency of plant population by using Quadrature method-B' 6 M
3. Identify the given plant material, write the characters and draw neat labelled diagram- 'C' (Hydrophytes /Xerophytes) 3 M
4. Identify and write about the morphology and medicinal uses of plant ' D&E' used in primary health care (2x1 =2) 2M
5. Identify and write about the morphology, active principles, uses of plant 'F&G' used in primary health care traditional medicine. (2x1=2) 2M
6. Herbarium 2M
7. Record and Viva 3 M

  
**Dr. P. KARUNAKAR RAO**  
Chairperson  
Board of Studies in Botany  
Kakatiya University  
Warangal, Telangana-506009, WARANGAL-506005 I.I.C

**Head**  
Dept. of Botany, Kakatiya University  
Warangal, Telangana-506009

## **B.Sc. BOTANY**

**II YEAR: Semester-IV**

### **Paper IV: Cell Biology, Genetics and Plant Physiology**

**DSC-1D (4 hrs./week)**

**Credits-4**

#### **Theory Syllabus (60 hours)**

##### **UNIT I:**

(15h)

1. Plant cell envelopes: Ultra structure and function of cell wall. Cell membrane and various models of cell membrane.
2. Cell Organelles: Structure and function of Nucleus, Mitochondria and Chloroplast.
3. Structure and function of DNA and RNA. Types of DNA and RNA. Chromosomes: Morphology, organization of DNA in a chromosome (Nucleosome model); Euchromatin and Heterochromatin.
4. Cell division: Mitosis and Meiosis and significance.

##### **UNIT II:**

(15h)

5. Mendel's Laws of inheritance, Incomplete dominance and Codominance. Test cross and Back cross.
6. Linkage- definition, types, Linkage maps: Two point and three point test cross.
7. Crossing over- mechanism and significance.
8. Mutations: Definition. Chromosomal aberrations (structural and numerical). Gene mutations (Frame Shift and Substitution mutations).

##### **UNIT-III**

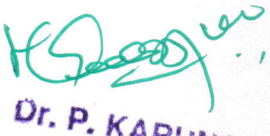
(15h)

9. Plant -Water Relations: Physical properties of water, diffusion, Osmosis; Osmotic potential. Absorption of water (Active and Passive)
10. Mineral Nutrition: Essential macro and micro mineral nutrients and their deficiency symptoms.
11. Transpiration; Structure of Stomata, Mechanism of transpiration; Translocation: Mechanism of Phloem transport (Source and Sink relationship)
12. Photosynthesis: Photosynthetic pigments, Photosystems (I and II), Photophosphorylation. (Cyclic and Noncyclic) Carbon assimilation pathways: Calvin cycle (C3), Hatch and Slack cycle (C4) Crassulacean Acid Metabolism (CAM)

##### **UNIT- IV**

(15h)

13. Respiration: Aerobic and Anaerobic; Glycolysis, Krebs cycle and electron transport system.
16. Enzymes: Nomenclature and Characteristics. Coenzymes and Co-factors, Enzyme mechanism.
17. Nitrogen Metabolism: Biological nitrogen fixation. Biochemistry of nitrogen fixation.
18. Physiological effects of Phytohormones: Auxins, Gibberellins, Cytokinins, ABA and Ethylene.

  
  
  
  
**Dr. P. KARUNAKAR RAO**  
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Dept. of Botany, Kakatiya University,  
Warangal, Telangana-506009.

### Suggested Readings:

1. Sharma, A. K. and A. Sharma. 1999. Plant Chromosomes: Analysis, Manipulation and Engineering. Harward Academic Publishers, Australia.
2. Shukla, R. S. and P. S. Chandel. 2007. Cytogenetics, Evolution, Biostatistics and Plant Breeding. S.Chand & Company Ltd., New Delhi.
3. Verma, P. S. and V. K. Agrawal. 2004. Cell Biology, Genetics, Molecular Biology, Evolution and Ecology. S. Chand & Company Ltd., New Delhi.
1. Hopkins, W. G. 1995. Introduction to Plant Physiology. John Wiley & Sons Inc., New York, USA
5. Jain, J.L., S. Jain and Nitin Jain. 2008. Fundamentals of Biochemistry. S. Chand & Company Ltd., New Delhi.
6. Pandey, B. P. 2007. Botany for Degree Students: Plant Physiology, Biochemistry, Biotechnology, Ecology and Utilization of Plants. S. Chand & Company Ltd., New Delhi.
7. Salisbury, F. B. and C. W. Ross. 1992. Plant Physiology. 4th edn. (India Edition), Wordsworth, Thomson Learning Inc., USA.
8. Taiz, L. and E. Zeiger. 1998. Plant Physiology (2nd Ed.). Sinauer Associates, Inc., Publishers, Massachusetts, USA.
9. Dutta A.C. 2016. Botany for Degree Students. Oxford University Press.
10. P.S. Bajaj. 1986 to 1990. Biotechnology in Agriculture and Forestry. Vol. 1 to 16. Ed
11. Richard N. Artica. 1996. Plant Growth Substances Principles and application. Chapman and Hall Inc. NY ( Ind Ed. Cbs. Publ and Dist, New Delhi 1997)
12. H. N. Krishnamurty, 1993, Physiology of plant growth and development Atmaramand Sons, New Delhi
13. V.Verma. Text Book of Plant Physiology. Emkay Publications., B-19, East KrishnaNagar, Delhi-1100051.
14. Jain, V.K. (2000): Fundamentals of Plant Physiology, S.Chand and Co, New Delhi. 12. Pandey, S.N. (1991): Plant Physiology, Vikas Publishing House (P) Ltd., New Delhi, India.
15. Verma, V. (2007): Text Book of Plant Physiology. Ane Books India, New Delhi. 17



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Dr. P. KARUNAKAR RAO  
Chairperson

Board of Studies in Botany  
Kakatiya University,

WARANGAL-506006 (I.G) INDIA



**B.Sc. BOTANY**  
**II YEAR: Semester-IV**  
**Paper IV: Cell Biology, Genetics and Plant Physiology**  
**DSC-1D** **Credits-1**

**Practical Syllabus (45 hours)**

1. Demonstration of cytochemical methods: Fixation of plant material and nuclear staining for mitotic and meiotic studies.
2. Study of various stages of mitosis using cytological preparation of Onion root tips.
3. Study of various stages of meiosis using cytological preparation of Onion flower buds.
4. Study of ultra structure of cell organelles using photographs Chloroplast, Mitochondria, Nucleus,
5. Study of structure of DNA and types
6. Solving the problems on Mendel's Laws, Incomplete dominance, two point and three-point test cross.
7. Study of various Chromosomal aberrations and gene mutations.
8. Determination of osmotic potential of vacuolar sap by Plasmolytic method using leaves of *Rheodiscolor* / *Tradescantia*.
9. Determination of rate of transpiration using Cobalt chloride method
10. Determination of stomatal frequency using leaf epidermal peelings / impressions
11. Separation of chloroplast pigments using paper chromatography technique
12. Estimation of protein by Biurette method
13. Mineral deficiency- Detail study of Micronutrients and Macro nutrients
14. Identification of C3, C4 and CAM plants

**Practical Model Question Paper**

**Time: 3 hrs**

**Max. marks: 25**

- |                                                                                                                        |    |
|------------------------------------------------------------------------------------------------------------------------|----|
| 1. Prepare a cytological slide of given material 'A' and identify & describe any two stages with well labeled diagrams | 6M |
| 2. Genetic problems – 'B'                                                                                              | 4M |
| 3. Physiological Experiment – 'C'.                                                                                     | 6M |
| 4. Spotter from cytology and genetics -(D & E) <span style="margin-left: 20px;">(2x2=4)</span>                         | 4M |
| 5. Physiology - Identify and Comment - (F & G) <span style="margin-left: 20px;">(2x1=2)</span>                         | 2M |
| 6. Record and Viva                                                                                                     | 3M |

  
  
  
  
**Dr. P. KARUNAKAR RAO**  
Chairperson  
Board of Studies in Botany  
Kakatiya University,  
WARANGAL-506005  
**Head**  
Dept. of Botany, Kakatiya University  
Warangal, Telangana-506009

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**KAKATIYA UNIVERSITY - WARANGAL - TELANGANA**  
**UNDER GRADUATE COURSES (UNDER CBCS 2021 – 2022 ONWARDS)**  
**B.SC. BOTANY III YEAR**  
**SEMESTER – V**

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**PAPER – V: (A) BIODIVERSITY & CONSERVATION**  
**(DSE-1: ELECTIVE)**

|           |               |                                                    |
|-----------|---------------|----------------------------------------------------|
| Theory:   | 4 Hours/Week; | Credits: 4 Marks: 100 (Internal: 20; External: 80) |
| Practical | 3 Hours/Week  | Credits: 1 Marks: 25                               |

**UNIT – I**

1. Plant diversity and its scope: Genetic diversity, Species diversity, Plant diversity at the ecosystem level, Agro biodiversity and cultivated plant taxa, wild taxa.
2. Values and uses of Biodiversity: Ethical and aesthetic values, Precautionary principle, Methodologies for valuation, Uses of plants, Uses of microbes.

**UNIT-II**

3. Loss of Biodiversity: Loss of genetic diversity, Loss of species diversity, Loss of ecosystem diversity, Loss of agro biodiversity, Projected scenario for biodiversity loss.
4. Management of Plant Biodiversity: Organizations associated with biodiversity, management- Methodology for execution-IUCN, UNEP, UNESCO, WWF, NBPGR.
5. Biodiversity legislation and conservation, Biodiversity information management and communication.

**UNIT-III:**

7. Conservation of Biodiversity: Conservation of genetic diversity, species diversity and ecosystem Diversity
8. Principles of conservation : *In -situ* and *Ex-situ* conservation. Sacred groove, Botanical garden, Biosphere reserves, Sanctuaries, National parks (*In-situ*) and Tissue culture, Gene / seed / pollen banks and Cryopreservation (*Ex-situ*).

**UNIT-IV:**

9. Role of plants in relation to Human Welfare; Importance of forestry, their utilization and commercial aspects, Avenue trees, Ornamental plants of India.
10. Alcoholic beverages through ages. Fruits and nuts, Important fruit crops and their commercial importance. Wood and its uses.



### References:

1. Krishnamurthy, K.V. (2004). An Advanced Text Book of Biodiversity - Principles and Practices. Oxford and IBH Publications Co. Pvt. Ltd. New Delhi
2. Bharucha, E. 2005. Textbook of Environmental Studies for Undergraduate Courses. Universities Press (India) Private Limited, Hyderabad.
3. Odum, E. P. 1983. Basics of Ecology. Saunder's International Students Edition, Philadelphia.
4. Sharma, P. D. 1989. Elements of Ecology. Rastogi Publications, Meerut.

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**KAKATIYA UNIVERSITY - WARANGAL - TELANGANA**  
**UNDER GRADUATE COURSES (UNDER CBCS 2021 – 2022 ONWARDS)**  
**B.SC. BOTANY III YEAR**  
**SEMESTER – V**

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**PAPER – V:: (A) BIODIVERSITY & CONSERVATION PRACTICAL**  
**(DSE-1: ELECTIVE)**

1. Study on local biodiversity: Herbs, shrubs and trees; Seasonal, Annual, biennial and perennial plants.
2. Study of morphological characteristics of plant communities: Hydrophytes (*Eichhornia*, *Hydrilla*, *Pistia*, *Nymphaea*, *Vallisneria*), Xerophytes: (*Asparagus*, *Opuntia*, *Euphorbia milii*, *Casuarina*, *Calotropis*).
3. Assessment of biodiversity
  - i) Avenue trees: *Pongamiapinnata*, *Butea monosperma*, *Spathodea* sp., *Delonix regia*, *Jacaranda mimosifolia*, *Cassia fistula*, *Mimusopselengi*, *Acacia leucophloea*, and *Albizialebbeck*.
  - ii) Ornamental Plants: Any five locally available ornamental plants.
  - iii) Timber Value: *Acacianilotica*, *Tectonagrandis* and *Azardirachta indica*
  - iv) Fruits: *Mangifera indica* (Mango), *Ziziphus mauritiana*, *Psidium guajava* (Guava), *Annona squamosa*
  - v) Nuts: *Anacardium occidentale* (Cashew), *Terminalia catappa* (Badam)
  - vi) Beverages: *Madhuca indica*, *Camellia sinensis* (Tea), *Coffea arabica* (Coffee), *Borassus flabellifer* (Toddy palm) and *Caryotaurens*
  - vii Medicinal value: *Catharanthus roseus*, *Tinosporacordifolia* and *Phyllanthus emblica*, *Ocimum* sp., and *Azardirachta indica*
4. Field trip: Collection of plants from the field, identification and preparation of Herbarium.

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**KAKATIYA UNIVERSITY - WARANGAL - TELANGANA**  
**UNDER GRADUATE COURSES (UNDER CBCS 2021 – 2022 ONWARDS)**  
**B.SC. BOTANY III YEAR**  
**SEMESTER – V**

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**PAPER – V: (B) ECONOMIC BOTANY**  
**(DSE-1: ELECTIVE)**

|           |               |                                                    |
|-----------|---------------|----------------------------------------------------|
| Theory:   | 4 Hours/Week; | Credits: 4 Marks: 100 (Internal: 20; External: 80) |
| Practical | 3 Hours/Week; | Credits: 1 Marks: 25                               |

**UNIT - I**

Origin of Cultivated Plants: Major plants introduction, Crop domestication and examples of crops / varieties

1. Vegetables: Nutritional and Commercial values of root crops, leafy and fruit vegetables.
2. Millets: Nutrient significance of Sorghum, Finger millet, Pearl millet, Foxtail millet.
3. Cereals: Rice, Wheat and maize - Origin, morphology and uses.

**UNIT – II**

4. Legumes: General account, importance to man and ecosystem.
5. Fruits and nuts: Commercial and nutritional value of South Indian fruits. Cashew nut, Almond and Walnut.
6. Sugars & Starches: Morphology and processing of sugarcane, products and by-products of sugarcane industry. Potato – morphology, propagation & uses.
7. Spices: Listing of important spices, part used, economic importance with special reference to fennel, saffron, clove and black pepper

**UNIT – III**

8. Beverages: Tea, Coffee (morphology, processing & uses)
9. Edible oils & Fats: General description, extraction, uses and health implications of groundnut, sunflower, coconut, linseed, and mustard.
10. Essential Oils: General account, extraction methods, comparison with fatty oils & their uses.
11. Natural Rubber: Para-rubber - tapping, processing and uses.

**UNIT – IV**

12. Drug-yielding plants: Therapeutic and habit-forming drugs with special reference to *Cinchona*, *Digitalis*, *Papaver* and *Cannabis*.
13. Tobacco processing, uses and health hazards
14. Timber plants: General account with special reference to teak and pine
15. Fibres: Classification based on the origin of fibres, extraction methods and uses of Cotton and Jute.

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### **Suggested Readings**

1. Kochhar, S.L. (2012). Economic Botany in Tropics, MacMillan & Co. New Delhi, India.
2. Wickens, G.E. (2001). Economic Botany: Principles & Practices. Kluwer Academic Publishers, The Netherlands.
3. Chrispeels, M.J. and Sadava, D.E. (2003). Plants, Genes and Agriculture. Jones & Bartlett Publishers.
4. B.P. Pandey (2007). Economic Botany, S. Chand & Company Ltd. New Delhi. 17/e.

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**KAKATIYA UNIVERSITY - WARANGAL - TELANGANA**  
**UNDER GRADUATE COURSES (UNDER CBCS 2021 – 2022 ONWARDS)**  
**B.SC. BOTANY III YEAR**  
**SEMESTER – V**

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**PAPER – V:: (B) ECONOMIC BOTANY PRACTICAL**  
**(DSE-1: ELECTIVE)**

1. Study of economically important plants: Wheat, Gram, Soybean, Black pepper, Clove Tea, Cotton, Groundnut through specimens, sections and microchemical tests.
2. Identification and study on nutrient values of locally available vegetables, millets and cereals.
3. Study on nutrient values and commercial status of Cashew nut, Almond and Walnut.
4. Uses and health implications of groundnut, sunflower, coconut, linseed, Brassica and Coconut.
5. Study of economically important plants : Wheat, Gram, Soybean, Black pepper, Clove Tea, Cotton, Groundnut through specimens, sections and microchemical tests
6. Study of products of economic importance included unit wise.
7. Collection vegetable twigs and preparation of Herbarium.
8. Identification of starch granules.
9. Estimation of iodine number of different oils.
10. Quantitative estimation and comparative study of proteins in millets and cereals.

B. Lalitha Prasad Sharma AF

**KAKATIYA UNIVERSITY - WARANGAL - TELANGANA**  
**UNDER GRADUATE COURSES (UNDER CBCS 2021 – 2022 ONWARDS)**  
**B.SC. BOTANY III YEAR**  
**SEMESTER – V**

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**PAPER – V:: (C) SEED TECHNOLOGY**  
**(DSE-1: ELECTIVE)**

|           |               |                                                    |
|-----------|---------------|----------------------------------------------------|
| Theory:   | 4 Hours/Week; | Credits: 4 Marks: 100 (Internal: 20; External: 80) |
| Practical | 3 Hours/Week; | Credits: 1 Marks: 25                               |

**UNIT- I**

1. Seed: Structure and types. Seed development in cultivated plants, seed quality concept, importance of genetic purity of seed. Hybrid seed production and Heterosis.
2. Cross pollination, Emasculation, role of pollinators and their management.
3. Collection and storage of pollen for artificial pollination.

**UNIT-II**

4. Seed germination: Internal and external factors affecting germination.
5. Physiological processes during seed germination; seed respiration, breakdown and mobilization of stored seed reserves.
6. Seed dormancy: Types, causes and methods of breaking dormancy. Role of Phytochrome.

**UNIT-III**

7. Cultural practices and harvesting of Seed: Isolation, Sowing, Cultural practices, harvesting and threshing of the following crops: a) Rice b) Cotton c) Sunflower
8. Seed treatment to control seed borne disease –General account
9. Seed testing- Procedures of seed testing, seed testing laboratories and importance of seed testing.

**UNIT-IV**

10. Seed viability, factors affecting seed viability and genetic erosion.
11. Seed storage: Long term and short term storage. Orthodox and recalcitrant seeds.  
Packing of seeds – Principles, practices, bagging and labelling.
12. Seed banks- National, International and Millennium seed banks. Seed certification- History, Seed certification agency, Indian millennium, general and specific seed certification standard.

*C. Lalitha Rao* *Sharma* *al*

## Reference:

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B. Lalitha K. Ram S. Sharma S. f.

**KAKATIYA UNIVERSITY**  
**U.G. BOTANY (Under CBCS)**  
**B.Sc. Final Year (DSE-1F)**  
**SEMESTER – VI**

**Elective I**

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**A) Tissue Culture and Biotechnology**

**DSE-1F (3 hrs./week)**

**Theory Syllabus**

**Credits-3**  
**(45 hours)**

**Unit – I**

1. Tissue culture: Introduction, sterilization procedures, explants, culture media – composition and preparation; Micropropagation. (5h)
2. Organ culture: Vegetative Organs-Root, Shoot, Leaf culture (6h)  
Reproductive Organs-Anther, Ovary, Ovule, Embryo culture
3. Callus culture, Cell and Protoplast culture (4h)

**Unit – II**

4. Somatic hybrids and Cybrids. (4h)
5. Applications of tissue culture: Production of pathogen free plants and somaclonal variants, production of stress resistance plants, secondary metabolites and synthetic seeds. (6h)
6. Production of hairy roots and its applications in production of secondary metabolites. (2h)

**Unit – III**

7. Biotechnology: Introduction, history, scope and applications. (3h)
8. rDNA technology: Basic aspect of of gene cloning, Enzymes used in gene cloning – Restriction enzymes, Ligases, Polymerases. (4h)
9. Gene cloning-Vectors – cloning vehicles (Plasmid, Cosmids, Bacteriophages, & Phasmids) application of r DNA technology. (5h)

**Unit – IV**

10. Gene Libraries: Genomic Libraries, cDNA Libraries, Polymerase chain reaction and its applications. (4h)
11. Method of gene transfer in plants (*Agrobacterium* and Microprojectile) (4h)
12. Production of transgenic plants, Bt –application in cotton and brinjal. Application of Transgenic in crop improvement. (3h)

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**B.Sc. Final Year (DSE-1F)**  
**SEMESTER – VI**

**Elective I**

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**A) Tissue Culture and Biotechnology**  
**Practical Syllabus**

- |                                                                                                     |       |
|-----------------------------------------------------------------------------------------------------|-------|
| 1. Estimation of plant DNA. (Tomato)                                                                | (6h)  |
| 2. Production of synthetic seeds /Encapsulation of embryo                                           | (3 h) |
| 3. Preparation of plant tissue culture medium.                                                      | (6h)  |
| 4. Callus Micropropagation                                                                          | (3h)  |
| 5. Demonstration of Micropropagation/ multiple shoots                                               | (6h)  |
| 6. Anther culture                                                                                   | (3 h) |
| 7. PCR –Demonstration                                                                               | (3h)  |
| 8. Study of biotechnology products: Samples of antibiotics and vaccines                             | (6h)  |
| 9. Photographs of transgenic plants – Bt Cotton, Bt –Brinjal.                                       | (3h)  |
| 10. Instruments used in Biotechnology lab- Autoclave, Laminar air flow, Hot air oven and Incubator. | (6h)  |